

Hitha Shri Nagaruru

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F-1 student at UCI (MS CS, Dec 2026). CPT-eligible for internship roles. No sponsorship required for internship. Seeking AI/ML Engineering internship for Summer/Fall 2026.

EDUCATION

University of California, Irvine

Sep 2025 – Dec 2026

Master of Computer Science | GPA: 3.83/4.00

JSS Academy of Technical Education, Bengaluru

Dec 2020 – Jun 2024

BE in Computer Science and Engineering | GPA: 3.81/4.00

EXPERIENCE

Bosch Limited — AI Engineer

Aug 2024 – Aug 2025

Digitization & Industry 4.0 — full-time industry role prior to MS

- Built a **production RAG chatbot** (GPT-4o + FastAPI) for real-time MES breakdown resolution — 96% accuracy on known failure cases, 100% safe fallback for unseen errors, zero manual input required.
- Developed a **Daily Leadership Review chatbot** using Azure OpenAI + FastAPI integrated with live Oracle database, eliminating 150+ Power BI licenses and delivering real-time KPI insights to plant leadership.
- Deployed the first **Python-based on-premise OCR web app** within the plant network (Flask + Waitress/NSSM) for secure HR document digitization — no external data exposure.
- Designed an **automated OCR data pipeline** (Tesseract, PyMuPDF, Pandas) that converted machine-generated PDFs from shared network storage into structured Excel datasets feeding live Power BI dashboards.

PROJECTS

Federated Learning Engine with Privacy Risk Detection

Mar 2026 – Present

- Building a **federated learning system** with non-IID data distribution across simulated clients and FedAvg aggregation; implementing membership inference attacks to detect training data leakage.
- Integrating **differential privacy mechanisms** (gradient clipping, noise injection) into training pipeline; benchmarking privacy–utility tradeoffs across varying noise levels.

LLM Validation Framework

Jan 2026 – Present

- Engineered a **modular validation framework** with fail-fast sequential pipeline (toxicity → accuracy), calibrated 0–1 scoring, and early termination on unsafe or ungrounded model outputs.
- Built a **3-layer toxicity detection system**: rule-based filtering + Detoxify scoring + SentenceTransformer semantic similarity (all-MiniLM-L6-v2) across configurable policy categories.
- Developed accuracy agent using **LLM-as-a-judge** via DeepEval (GEval) + Anthropic Claude Haiku with BM25-ranked web retrieval for factual grounding.

SKILLS

Generative AI & LLMs: Azure OpenAI (GPT-4o), RAG, Prompt Engineering, FastAPI, LLM Evaluation (DeepEval, GEval), Anthropic Claude, Hugging Face

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Federated Learning, Differential Privacy, Anomaly Detection, Hypothesis Testing

Languages & Infra: Python, SQL (PostgreSQL, Oracle, MySQL), PySpark, Databricks, Azure ML Studio, Flask, Git, CUDA